

Hodge Conjecture: The Geometric Inevitability Depicted by Mathematical Regulations

[Original Problem]

"Is every Hodge class on a non-singular complex projective variety

a rational linear combination of algebraic cycles?"

This question asks whether anything that can be depicted as a "shape (geometric extent)" and that satisfies certain conditions can necessarily be constructed using "numbers (algebraic expressions)."

1. Intermediate development of synchronization: Forms that deviate from the regulations

Step A: The "shape" of a geometric

manifold does not exist independently, but is the "interference fringe" itself drawn by the underlying mathematical rules.

$$\text{Geometry} = \text{Inference Interference}$$

Geometry is the interference pattern of underlying mathematical definitions.

Step B: Synchronization by mathematical ratios

As long as the ratio "1 is 0.5 of 2" is maintained mathematically, there is no room for discrepancy between algebraic "numbers" and geometric "shapes".

$$[Z] \cap [H]^\perp = 0$$

As long as the ratio "1 is 0.5 of 2" is maintained, no discrepancy exists between numbers and shapes.

Step C: Necessary Agreement (Synchronization)

The agreement of Hodge classes (shape features) with algebraic cycles (numerical constructions) is a necessary consequence from which the results leak out of the definition.

Once the blueprint (rules) is decided, the final design (shape) is essentially already written into it from the beginning.

Synchronization Perfected

The alignment is a necessary consequence; once the rules are set, the final form is already inherent within them.

Critique of modern mathematics: The illusion of separation

The fallacy of modern algebraic geometry lies in **treating "numbers (algebra)" and "shape (geometry)" as separate domains and wondering "why they coincide . "**

- **The division of essence:** Mathematicians consider "blueprints" and "buildings" to be separate things, but in a consistent mathematical space, the two are inseparable and identical.
- **Overconfidence in "proof":** They pile up complex calculations, but all they are doing is desperately reaffirming that "the building will be constructed according to the blueprints."
- **The forgetting of the absence of inconsistency:** If inconsistency is eliminated in mathematical definitions in the first place, then it would be impossible for form and number to diverge from the beginning.

Calling the Hodge conjecture a difficult problem is nothing more than evidence of being trapped in a labyrinth of abstraction of one's own making, and losing sight of the intuitive truth that **"numbers define form, and form is number itself."**

Modern mathematics errs by separating algebra from geometry, failing to see them as identical manifestations of the same underlying logic.